AI-Driven Equity Crowdfunding Platform

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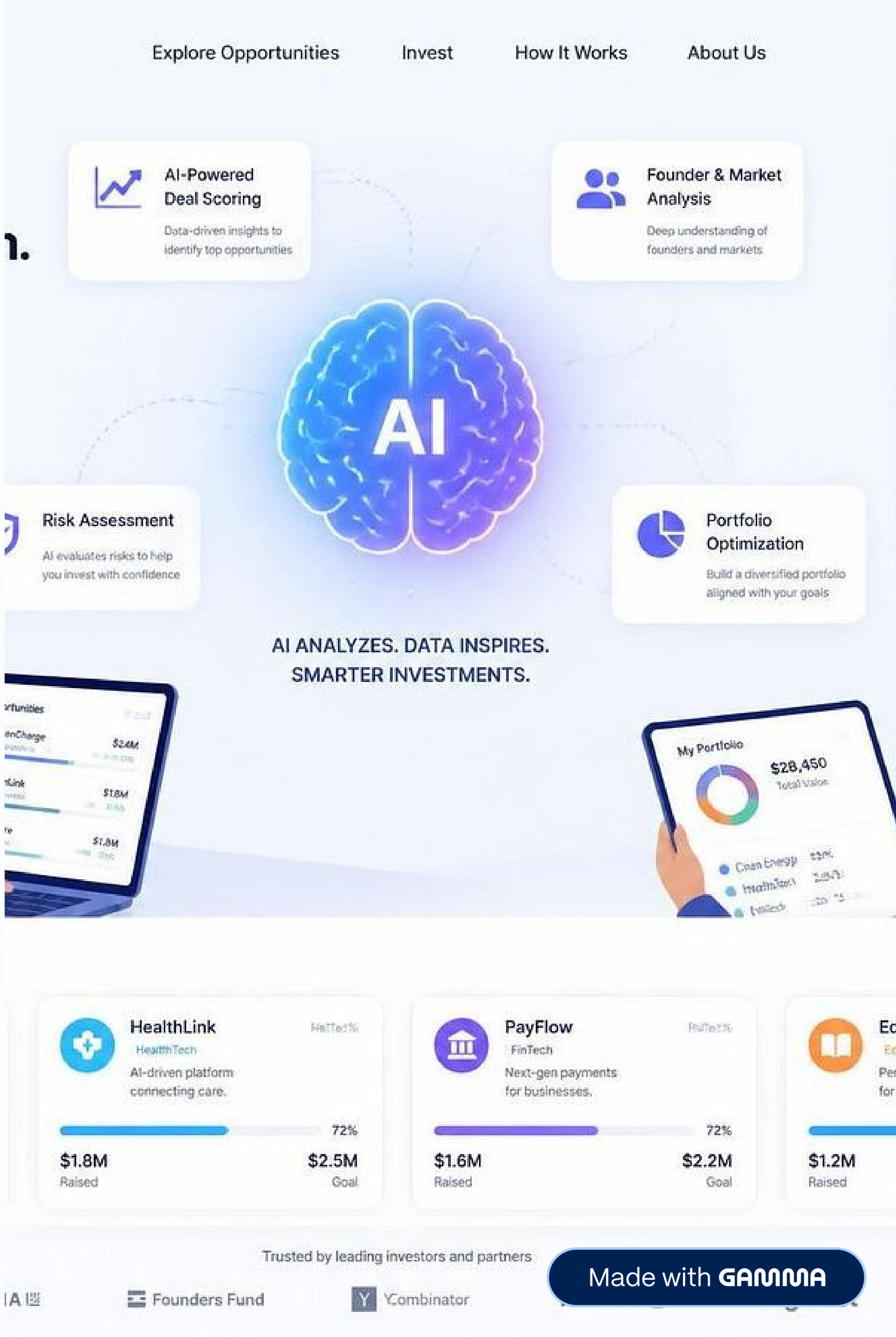


Problem & Proposed Solution

Equity crowdfunding connects startups with investors, but investors face:

1

Problems

- Information asymmetry
- Unstructured startup information
- Difficult startup evaluation
- Trust and verification issues
- Difficulty monitoring investments

2

AI-Driven Equity Crowdfunding Platform

Bridges the gap with intelligent, structured support.

3

Solutions

- Structured Startup Information
- AI-Based Evaluation
- Explainable Insights
- Verification & Security
- Investment & Portfolio Monitoring

AI assists investors; it does not make the final investment decision.

Objectives & Relevance

Project Objectives

Platform

Startup registration, fundraising campaigns, structured information

AI

ML-based analysis, explainable insights

Trust & Security

User/startup verification, JWT + Spring Security + RBAC

Investment

Portfolio management, campaign and valuation tracking

Industry & Societal Relevance

FinTech

Digital startup fundraising

AI & Data Analytics

Data-driven evaluation

Cybersecurity

Secure platform access

Entrepreneurship

Alternative funding channel

Literature Survey

Research provides individual building blocks—evaluation, trust, behaviour, ML and XAI—that our project integrates into one platform.

Paper	Method / Dataset	Key Finding	Relevance to Project
Ralcheva & Roosenboom (2020)	Logistic Regression; 2,171 campaigns	Campaign characteristics help forecast success.	Basis for startup & campaign attributes for AI evaluation.
Ferrati & Muffatto (2021)	Review; 53 studies	Investors use diverse financial & non-financial criteria.	Identifies information needed for startup evaluation.
Mazzocchini & Lucarelli (2023)	SLR; 32 studies	Success depends on information, campaign, investor & trust factors.	Supports structured information, transparency & verification.
Sendra-Pons et al. (2024)	QCA; Startupxplore	Presentation & social signals influence trust.	Supports transparent & credible startup profiles.
Estrin et al. (2024)	Cross-platform; 1,872 observations	Examines investor participation and funds raised.	Supports the digital fundraising model.
Åstebro et al. (2024)	Empirical; 69,699 pledges	Investment activity influences later behaviour.	Supports campaign & investment activity visibility.
Yang et al. (2025/26)	ML + SHAP; 144 campaigns	ML identifies influential cues; SHAP explains them.	Direct foundation for ML + Explainable AI.
Joshipura et al. (2026)	Bibliometric; 528 articles	ECF research spans success, investors, trust & platforms.	Provides the broader research foundation.

Research Gap & Research Opportunity

Gaps Identified from Existing Research

1. **Fragmented Research:** Campaign success, investor criteria, trust, investor behaviour, ML and XAI are mostly studied as separate areas.
2. **Prediction ≠ Explanation:** ML can predict outcomes, but investors also need to understand which factors influence the prediction and why.
3. **Research ≠ Platform:** Existing ML/XAI studies focus mainly on analysis and prediction rather than integrating these techniques into a complete crowdfunding platform.
4. **Limited End-to-End Integration:** AI analysis is rarely combined with startup verification, secure investment management and portfolio monitoring in one system.
5. **Investor-Centric Gap:** Research findings are not always converted into simple, actionable insights within the investor's actual decision workflow.

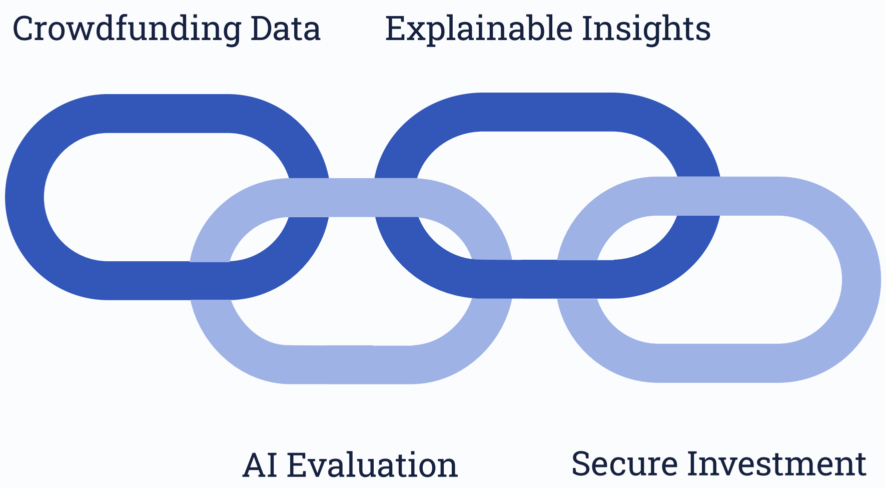
Key Opportunity

> From isolated research components → to an integrated AI-driven crowdfunding ecosystem

Our Research Opportunity

Bridge the gap between crowdfunding research and practical investor decision support by integrating ML, Explainable AI, verification, security and investment monitoring into one unified platform.

Proposed Research Direction

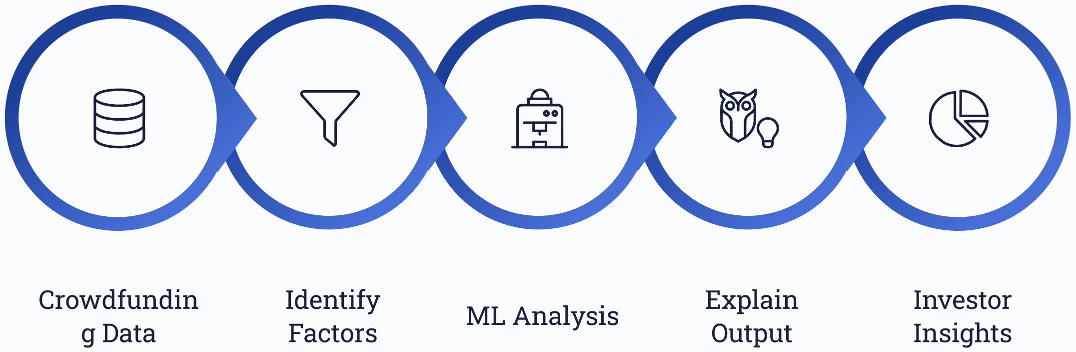


Research Focus & Analytical Framework

What the Project Investigates

Research Area	What We Examine
Startup Evaluation	Which startup and campaign factors are relevant for evaluating an investment opportunity?
Predictive Analysis	Can machine learning identify meaningful patterns in crowdfunding data?
Explainable AI	Which factors influence the model's assessment, and how can they be explained to investors?
Platform Integration	How can AI insights be incorporated into the investor workflow without replacing human judgment?
Continuous Monitoring	How can campaign activity, investments and portfolio information be tracked effectively?

Research → Project



Key Research Focus

Evaluate → Analyze → Explain → Integrate → Monitor

Project Development Methodology

1. **Requirement Analysis**
Identify requirements of Founders, Investors and Admins.

2. **System Design**
Design roles, database, modules, AI workflow and security.

3. **Data Preparation**
Collect, clean and preprocess crowdfunding data.

4. **AI Model Development**
Train and compare suitable ML models.

5. **Explainable AI**
Use an approach such as SHAP to explain model outputs.

6. **Platform Development**
Develop authentication, campaigns, investments, portfolios and verification.

7. **AI–Platform Integration**
Connect the trained model with the crowdfunding platform.

8. **System Validation**
Validate functionality, AI output, security, usability and integration.

Flow

Requirements → Design → Data → AI → Platform → Integration → Validation

AI & Explainable AI

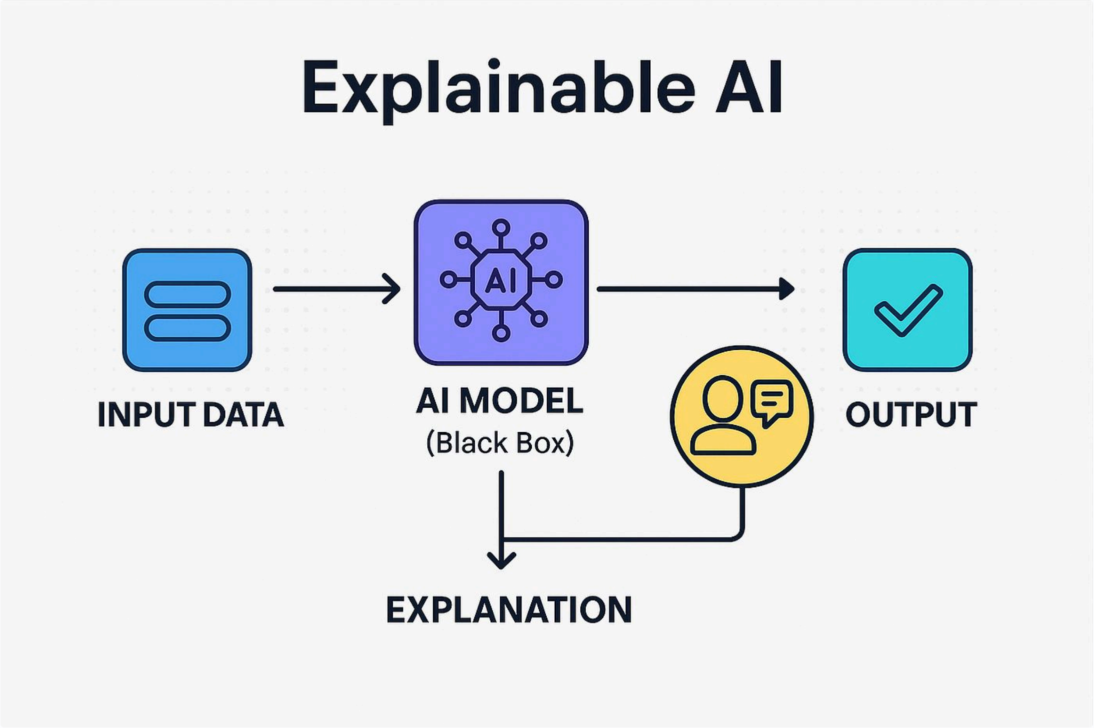
AI Processing Pipeline

1 **Crowdfunding Data**
Raw campaign & startup inputs

2 **Preprocessing**
Cleaning, feature engineering

3 **ML Model**
Prediction & evaluation

4 **SHAP / XAI**
Investor-friendly insights



Why XAI?

- A prediction alone does not explain why it was generated
- XAI identifies important influencing factors, feature contributions, and reasons behind model output
- Goal: Make AI-based evaluation understandable and transparent for investors

Platform Architecture

Conclusion

The project brings research-backed crowdfunding analysis and AI-based decision support together within a secure, transparent and investor-focused platform.

What the Literature Shows

Equity crowdfunding requires consideration of evaluation, information, trust, behaviour, and monitoring. Research demonstrates the potential of **Machine Learning + Explainable AI**.

Our Proposed Direction

AI-Driven Equity Crowdfunding Platform integrating crowdfunding, AI/ML, explainability, security, verification, and investment monitoring.

References

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